MEDICAL CORE · VERIFIED RESEARCH

Mandibular Fracture Reduction Optimization

Convert an ambiguous clinical problem into a testable 3D registration and optimization problem.

EVIDENCE STATE	PERIOD
Verified · Completed	2021.12 - 2023.02
PORTFOLIO TIER	**PRIMARY CAPABILITY**
Medical Core	3D Geometry & Registration

Built a surgical-planning simulator and experiment pipeline that optimizes mandibular fragment pose using dental features and fracture surfaces.

Problem and system boundary

01 / PROBLEM DEFINITION

Problem definition

Defined occlusion as a reproducible target for fracture reduction.

PROBLEM

Many geometrically plausible reductions existed; the planning target needed a reproducible constraint grounded in occlusion.

Built a surgical-planning simulator and experiment pipeline that optimizes mandibular fragment pose using dental features and fracture surfaces.

PUBLIC BOUNDARY

Results are from prepared research datasets; this does not claim routine clinical use or sole first authorship.

Clinical interpretation, algorithm design, experiment design, and writing were conducted as joint research.

EVIDENCE STATE

Verified · Completed / Presentation and award media remains pending approval; the publication is publicly referenced.

Owned decisions and implementation

MY ROLE

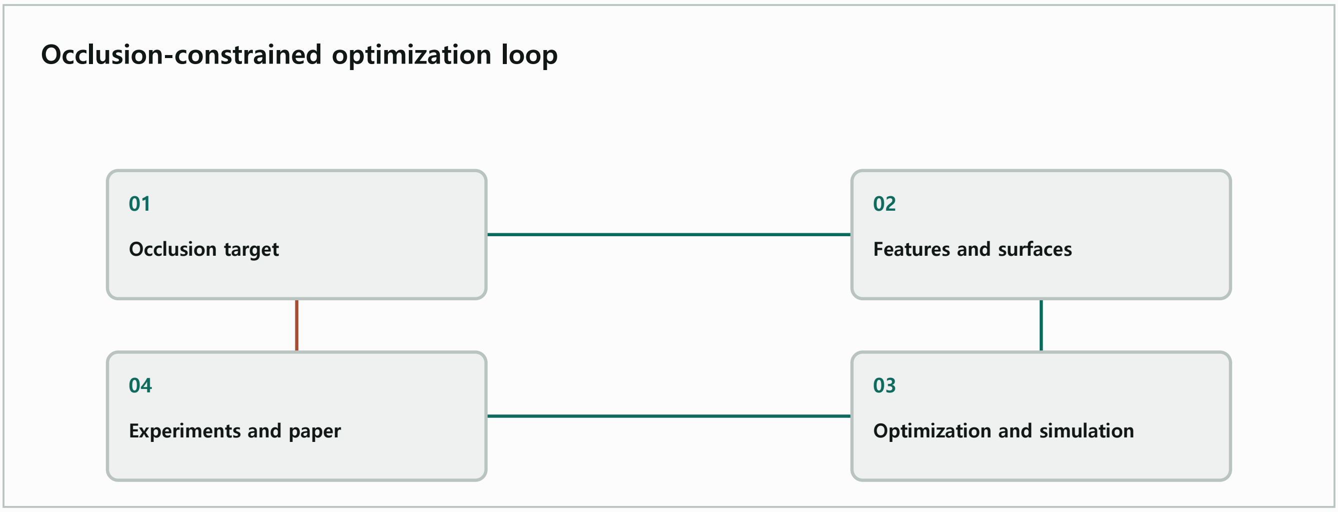
Jointly led the research pipeline across problem definition, feature and optimization design, simulator, experiments and analysis, and paper, serving as a co-first author.

02 / RESEARCH PIPELINE

Research pipeline

Feature design · Optimization · Simulator · Experiments and analysis · Paper

SYSTEM RELATIONSHIP



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

Python / Open3D / OpenCV / SciPy / PCA / ICP / CLPSO / 3D Slicer

Verification and evidence ledger

03 / PUBLISHED EVIDENCE

Published evidence

Separate presentations, award, quantitative experiments, and the peer-reviewed publication.

REGISTERED PUBLIC EVIDENCE

IMAGE / PENDING PUBLIC APPROVAL

mandibular-presentation-award

Presentation or recognition evidence; no public derivative approved.

PUBLICATION / PUBLICLY INSPECTABLE

mandibular-publication

Open public link

Public article page used as external research evidence.

VERIFICATION AND EVIDENCE LEDGER

ACCAS 2022, a 2022 domestic conference, a 2023 best-paper award, a 2024 Q1 SCIE peer-reviewed paper, and quantitative experiments provide the evidence.

Role, team result, and limits

04 / RESEARCH BOUNDARY

Research boundary

Prepared-dataset results do not establish routine clinical use.

MY ROLE

Jointly led the research pipeline and served as a co-first author.

Jointly led the research pipeline across problem definition, feature and optimization design, simulator, experiments and analysis, and paper, serving as a co-first author.

TEAM RESULT

The joint research team produced international and domestic presentations, an award, and a peer-reviewed publication.

ACCAS 2022, a 2022 domestic conference, a 2023 best-paper award, a 2024 Q1 SCIE peer-reviewed paper, and quantitative experiments provide the evidence.

LIMITATIONS

Results are from prepared research datasets; this does not claim routine clinical use or sole first authorship.

Presentation and award media remains pending approval; the publication is publicly referenced.

COLLABORATION BOUNDARY

Clinical interpretation, algorithm design, experiment design, and writing were conducted as joint research.

Built a surgical-planning simulator and experiment pipeline that optimizes mandibular fragment pose using dental features and fracture surfaces.

Recap and joint-development invitation

Convert an ambiguous clinical problem into a testable 3D registration and optimization problem.

PRIMARY CAPABILITY

3D Geometry & Registration / Medical Navigation & Visualization

REGISTERED PUBLIC EVIDENCE

Publication

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

Clinical interpretation, algorithm design, experiment design, and writing were conducted as joint research.

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